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Chapter 1. Introduction

1.1 Information

Information

Information refers to data or knowledge that is meaningful and has value in making decisions or solving problems.

1.1.1 Importance of Information Theory

1.1.1.1 Why Information Theory Matters

- It helps us **quantify** information.
- It helps us **transmit** information efficiently.
- It helps us **optimize the delivery** of information in communication, computing, and biological systems.

1.1.1.2 Real-Life Relevance

- Modern systems such as the **internet**, **mobile networks**, and **social media** rely on efficient data compression and error correction.
- These ideas ensure messages are transmitted **accurately** and **efficiently**.
- Information theory also helps reduce **noise interference** in signals.
- This improves reliability in applications such as **streaming**, **medical diagnostics**, and **genomic analysis**.

1.1.2 Importance in Communication

1.1.2.1 Key Points

- Information theory provides mathematical frameworks to optimize **data transmission** and **storage**.
- It helps determine the maximum amount of information a channel can handle, that is, **channel capacity**.
- It helps determine how to encode messages so that transmission errors can be reduced.

1.1.2.2 Communication Insight

- Cellular networks and internet protocols use **error correction** and **data compression**.
- These make communication **efficient** and **reliable**.
- Even in noisy environments, users can still experience:
  - clear calls,
  - fast internet speeds,

- accurate data exchange.

This is why information theory is fundamental to seamless global communication.

1.2 Discrete Source

Discrete Source

A discrete source is a system that generates a **finite set of symbols or values** from a **predefined set of alphabets**. These symbols are discrete and distinct, such as letters, numbers, or pixels.

1.2.0.1 Examples of Discrete Sources

- **Text:** A keyboard generates letters, numbers and special characters.
- **Audio:** Digital audio files sample sound at specific intervals, producing discrete data points.
- **Image/Video:** Images are composed of pixels with distinct color values, while videos are sequences of discrete image frames.

1.2.1 Source Types

1.2.1.1 Memoryless Source

Memoryless Source

A discrete source is memoryless if the symbols it generates are **independent** of each other.

- The current symbol has **no dependence** on past symbols.
- Example: flipping a fair coin repeatedly produces independent outcomes (H or T).
- Such sources simplify analysis because each symbol can be treated individually without considering historical context.
- Example:A random number generator producing digits (0–9) where each number is equally likely and independent of the previous numbers.

1.2.1.2 Memory Source

Memory Source

A discrete source is a memory source if the generation of a symbol **depends** on one or more **previous symbols**.

- Such sources have **statistical dependency** between symbols.
- This dependency can be modeled using **Markov processes** or **conditional probabilities**.
- They are common in **natural language and encoded data streams**, where rules and structure influence symbol generation.
- Example: In English text, the letter ‘q’ is almost always followed by ‘u’, showing a strong dependency between these symbols.

1.3 Self Information

Self Information

Self information measures the **amount of uncertainty or surprise** associated with the occurrence of a specific event. It quantifies how much **information is gained** when an event occurs.

Self Information Formula

Consider a discrete random variable X with possible outcomes x_i, $i = 1, 2, 3, \ldots, n$. Self information of the event $X = x_i$ is defined as

$$I(x_i) = \log \frac{1}{P(x_i)} = -\log P(x_i).$$

- ▹ $I(x_i)$ = self information of the event x_i
- ▹ $P(x_i)$ = probability of the event x_i

1.3.1 Properties of Self Information

1. If we are absolutely certain of the outcome of an event even before it occurs, there is no information gained.

$$I(x_i) = 0, \quad \text{if } P_i = 1$$

   This means **certainty carries no message information**.

2. The occurrence of an event either provides some or no information, but never brings about a loss of information.

$$I(x_i) \geq 0, \quad \text{for } 0 \leq P(x_i) \leq 1$$

   That implies **self information cannot be negative**.

3. The **less probable an event is, the more information we gain** when it occurs.

$$I(x_1) > I(x_2) \quad \text{if } P_1 < P_2$$

4. For statistically independent events,

$$I(x_1, x_2) = I(x_1) + I(x_2).$$

   Equivalently,

$$I_A + I_B = \log \frac{1}{P_A} + \log \frac{1}{P_B} = \log \frac{1}{P_A P_B} = \log \frac{1}{P(AB)}.$$

1.3.2 Physical significance / insight of self information

Consider a set of events $X = \{x_1, x_2, x_3\}$ with probability

$$P = \{0.5, 0.25, 0.25\}.$$

Step 1: Self information for the first event:

$$I_{x_1} = \log_2 \frac{1}{0.5} = -\log_2 0.5 = 1 \textbf{ bit}$$

Step 2: For the other two events:

$$I_{x_2} = I_{x_3} = \log_2 \frac{1}{0.25} = -\log_2 0.25 = 2 \textbf{ bit}$$

Observation

- **Higher probability requires fewer bits.**
- **Lower probability requires more bits.**
- **Lower probability therefore means a higher degree of uncertainty.**

1.4 Entropy

Entropy

In information theory, entropy is the measurement of the average amount of information, that is, the number of bits of uncertainty, in a set of possible outcomes per symbol.

1.4.1 Interpretation

- **It measures the unpredictability of a random variable.**
- **It represents the minimum number of bits required to encode the information generated by a source.**

Entropy of an Event

$$H(x_i) = -P_i \log_2 P_i.$$

- $H(x_i)$ = **entropy contribution of the event** x_i
- P_i = **probability of the** i**th event**

1.4.2 Why This Form Appears

- **Self information of an event is** $(-\log_2 P_i)$.
- **Entropy is an average information measure.**
- **So self information is weighted by its probability** (P_i).

Entropy of a Random Variable

$$H(X) = -\sum_{i=1}^{n} P_i \log_2 P_i \quad \textbf{bit/symbol.}$$

- $H(X)$ = **average entropy of the random variable** X
- n = **total number of possible outcomes**
- P_i = **probability of the** i**th outcome**

Example 1.1

A source generates information with probabilities

$$P_1 = 0.1, \quad P_2 = 0.2, \quad P_3 = 0.3, \quad P_4 = 0.4.$$

Find the entropy of the system and what percentage of maximum possible information is being generated by this source.

Step 1: Calculate entropy

$$H(X) = -\sum_{i=1}^{n} P_i \log_2 P_i$$

$$= -\Big(0.1\log_2(0.1) + 0.2\log_2(0.2) + 0.3\log_2(0.3) + 0.4\log_2(0.4)\Big)$$

$$= 1.8464 \approx 2 \textbf{ bits/symbol.}$$

Step 2: Calculate maximum entropy

$$H_{\max} = \log_2 4 = 2.$$

Step 3: Calculate efficiency

$$\eta = \frac{H}{H_{\max}} \times 100 = \frac{1.8464}{2} \times 100 = 92.32\%.$$

Question 1.2

Q: Why is P_i multiplied with self information in the entropy equation?
A:

- **Entropy is an *average* measure of information.**
- **So the self information term $(-\log_2 P_i)$ must be weighted by its probability P_i.**
- **This keeps highly probable events contributing more frequently and less probable events contributing less frequently to the final average.**
- **It also keeps the entropy value bounded properly for a valid probability distribution.**

1.4.3 Significance of Entropy

- **Measures uncertainty:** quantifies unpredictability in a source, providing insight into the information content.
- **Data compression:** sets the theoretical limits for lossless compression, helping design efficient encoding systems.
- **Error detection/correction:** guides the design of coding techniques to handle noise in communication channels.
- **System efficiency:** helps evaluate and optimize the performance of communication and storage systems.
- **Applications in AI/ML:** used in decision trees and deep learning for feature selection and model optimization.

1.4.4 Relation between Hartley, Nats and Bits

1.4.4.1 Units of Self Information

- **Self information in Hartley:**

$$I = \log_{10} \frac{1}{P}$$

- **Self information in nats:**

$$I = \log_{e} \frac{1}{P}$$

- **Self information in bits:**

$$I = \log_{2} \frac{1}{P}$$

Hartley to Bits

$$1 \textbf{ Hartley} = \frac{I}{-\log_{10} P} = \frac{-\log_2 P}{-\log_{10} P} = \frac{\log 10}{\log 2} = \log_2 10 = 3.3219 \textbf{ bits.}$$

Hartley to Nats

$$1 \textbf{ Hartley} = \frac{I}{-\log_{10} P} = \frac{-\log_e P}{-\log_{10} P} = \frac{\log 10}{\log e} = \log_e 10 = 2.302 \textbf{ nats.}$$

Nats to Bits

$$1 \textbf{ Nat} = \frac{I}{-\log_e P} = \frac{-\log_2 P}{-\log_e P} = \frac{\log e}{\log 2} = \log_2 e = \frac{1}{\ln 2} = 1.442 \textbf{ bits.}$$

Example 1.3

If there are 100 equiprobable outcomes, calculate the corresponding values in nats, bits and Hartley.
Given $N = 100$,

$$\textbf{In bits, } H(X) = \log_2(100) = 6.64$$
$$\textbf{In nats, } H(X) = \log_e(100) = 4.605$$
$$\textbf{In Hartley, } H(X) = \log_{10}(100) = 2$$

1.4.5 Equal Probability Leads to High Entropy

Consider a BPSK output

$$x = \begin{cases} 1, & \textbf{probability } p \\ 0, & \textbf{probability } 1-p \end{cases}$$

$$H(x) = -\sum_{x=0}^{1} P(x)\log_2 P(x)$$

$$= -p\log_2 p - (1-p)\log_2(1-p).$$

- p = **probability of symbol** 1
- $1-p$ = **probability of symbol** 0
- $H(x)$ = **entropy of the binary source**

  For equal probability $p = 0.5$ **and** $1-p = 0.5$**,**

$$H(x) = -0.5\log_2 0.5 - 0.5\log_2 0.5 = 1 \textbf{ bit/symbol.}$$

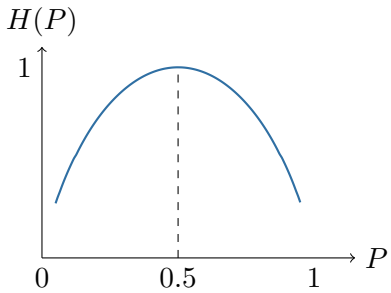

Figure 1.1: Entropy of a binary source versus probability

From the figure, it is clear that for $p = 0$ **or** $p = 1$**, entropy** $H(x)$ **is zero, but for** $p = 0.5$**, entropy is maximum. Equal probability leads to the highest uncertainty, so it also leads to the highest entropy.**

Example 1.4

Calculate the entropy of the QPSK transmitted signal below:

$$00 \;\; 01 \;\; 10 \;\; 10 \;\; 11 \;\; 11 \;\; 01 \;\; 01 \;\; 10 \;\; 00 \;\; 10 \;\; 10 \;\; 11 \;\; 01 \;\; 10$$

QPSK has four distinct $(00, 01, 10, 11)$ **bit combinations. Thus,** $2^n = 4 \Rightarrow n = 2$**. So, we have to take two bits for each symbol.**

Symbol	Times
00	**2**
01	**4**
10	**6**
11	**3**

$$P_{00} = \frac{2}{15}, \qquad P_{01} = \frac{4}{15}, \qquad P_{10} = \frac{6}{15}, \qquad P_{11} = \frac{3}{15}$$

$$\begin{aligned} H(X) &= -\left[P_{00}\log_2 P_{00} + P_{01}\log_2 P_{01} + P_{10}\log_2 P_{10} + P_{11}\log_2 P_{11}\right] \\ &= -\left[\frac{2}{15}\log_2\frac{2}{15} + \frac{4}{15}\log_2\frac{4}{15} + \frac{6}{15}\log_2\frac{6}{15} + \frac{3}{15}\log_2\frac{3}{15}\right] \\ &= 1.889 \approx 2 \textbf{ bits/symbol.} \end{aligned}$$

Physical significance: **Thus, we need at least 2 bits to transmit the signal.**

1.5 Joint Entropy

Joint Entropy

The joint entropy $H(X,Y)$ of a pair of random variables (X,Y) with joint distribution $P(x,y)$ is the average uncertainty associated with the pair of outcomes taken together.

Joint Entropy Formula

$$H(X,Y) = -\sum_{x\in X}\sum_{y\in Y} P(x,y)\log_2 P(x,y)$$
$$= -E\log_2 P(x,y)$$

- $H(X,Y)$ = **joint entropy of the variables** X **and** Y
- $P(x,y)$ = **joint probability of the pair** (x,y)
- $E[\cdot]$ = **expectation**

Example 1.5

Two variables are considered for a student:

- X = **study hours**
- Y = **attendance**

Their joint probabilities are listed below. Show that the variables are dependent on each other.

X	Y	P(X, Y)
Low	Low	0.20
Low	High	0.10
High	Low	0.15
High	High	0.55

Step 1: Joint entropy contribution of each case

$$H_{LL}(X,Y) = -0.2\log_2(0.2) = 0.46$$
$$H_{LH}(X,Y) = -0.1\log_2(0.1) = 0.33$$
$$H_{HL}(X,Y) = -0.15\log_2(0.15) = 0.41$$
$$H_{HH}(X,Y) = -0.55\log_2(0.55) = 0.4743$$

$$H(X,Y) = 0.46 + 0.33 + 0.41 + 0.4743 = 1.6743$$

Step 2: Marginal entropy for study hours

$$P_L = 0.2 + 0.1 = 0.3, \qquad P_H = 0.15 + 0.55 = 0.7$$

$$H(X) = -\left[0.3\log_2(0.3) + 0.7\log_2(0.7)\right] = 0.881$$

Step 3: Marginal entropy for attendance

$$P_L = 0.2 + 0.15 = 0.35, \qquad P_H = 0.1 + 0.55 = 0.65$$

$$H(Y) = -\left[0.35\log_2(0.35) + 0.65\log_2(0.65)\right] = 0.934$$

Conclusion

$$H(X) + H(Y) = 0.881 + 0.934 = 1.815$$

Since $H(X) + H(Y) > H(X,Y)$, the variables X and Y are dependent. We can also say that the variables contain mutual information.

1.6 Mutual Information

Mutual Information

Mutual information measures the amount of information that one random variable X provides about another random variable Y. It quantifies the reduction in uncertainty about one variable when the other is known.

Mutual Information Formula

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

or equivalently, $$I(X;Y) = H(X) - H(X \mid Y)$$

- $I(X;Y)$ = **mutual information between** X **and** Y
- $H(X)$, $H(Y)$ = **marginal entropies**
- $H(X,Y)$ = **joint entropy**
- $H(X \mid Y)$ = **conditional entropy of** X **given** Y

Example 1.6

Calculate the joint entropy for the following data and also find the mutual information.

Y \ X	**0**	**1**
0	$\frac{1}{2}$	$\frac{1}{4}$
1	$\frac{1}{8}$	$\frac{1}{8}$

Step 1: Joint entropy

$$\begin{aligned} H(X,Y) &= -\left[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + \frac{1}{8}\log_2\frac{1}{8} + \frac{1}{8}\log_2\frac{1}{8}\right] \\ &= 1.75 \end{aligned}$$

Step 2: Entropy of Y

$$P(Y=0) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}, \qquad P(Y=1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

$$H(Y) = \frac{3}{4}\log_2\frac{4}{3} + \frac{1}{4}\log_2 4 = 0.811$$

Step 3: Entropy of X

$$P(X=0) = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}, \qquad P(X=1) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$$

$$H(X) = \frac{5}{8}\log_2\frac{8}{5} + \frac{3}{8}\log_2\frac{8}{3} = 0.954$$

Step 4: Conditional entropies

$$H(X \mid Y) = H(X,Y) - H(Y) = 1.75 - 0.811 = 0.939$$

$$H(Y \mid X) = H(X,Y) - H(X) = 1.75 - 0.954 = 0.796$$

Step 5: Mutual information

$$I(X;Y) = H(X) + H(Y) - H(X,Y) = 0.954 + 0.811 - 1.75 = 0.015$$

1.7 Conditional Entropy

Conditional Entropy

Conditional entropy measures the amount of uncertainty remaining in a random variable when another related variable is already known. It tells us how much additional information is still needed to describe X when Y is already provided.

1.7.1 Simple Intuition

Consider the set $\{1, 2, 3, 4, 5, 6\}$. If we only know whether the outcome is odd or even, some uncertainty still remains.

- **If the outcome is odd, the possible values are $\{1, 3, 5\}$.**
- **If the outcome is even, the possible values are $\{2, 4, 6\}$.**

This remaining uncertainty is the idea captured by conditional entropy.

Conditional Entropy Formula

$$\begin{aligned} H(Y \mid X) &= -\sum_{x \in X} P(x) \sum_{y \in Y} P(y \mid x) \log_2 P(y \mid x) \\ &= -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(y \mid x) \end{aligned}$$

- $H(Y \mid X)$ **= uncertainty of** Y **after** X **is known**
- $P(y \mid x)$ **= conditional probability of** $Y = y$ **given** $X = x$
- $P(x, y)$ **= joint probability of** X **and** Y

1.7.2 Relation with Joint Entropy

Starting from the joint entropy,

$$\begin{aligned} H(X, Y) &= -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(x, y) \\ &= -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 \big(P(x) P(y \mid x)\big) \\ &= -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(x) - \sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(y \mid x) \\ &= -\sum_{x \in X} P(x) \log_2 P(x) - \sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(y \mid x) \end{aligned}$$

Hence,

$$H(X, Y) = H(X) + H(Y \mid X)$$

Similarly,

$$H(X, Y) = H(Y) + H(X \mid Y)$$

1.8 Relative Entropy

Relative Entropy

Relative entropy, also called Kullback-Leibler divergence, measures how different an estimated probability distribution is from the true distribution.

Kullback-Leibler Divergence

$$D(P \parallel Q) = \sum_{x \in X} P(x) \log_2 \frac{P(x)}{Q(x)}$$

- $P(x)$ = **true probability mass function**
- $Q(x)$ = **estimated probability mass function**
- $D(P \parallel Q)$ = **relative entropy from** Q **to** P

Basic Points

- **If** $P(x) = Q(x)$**, then** $D(P \parallel Q) = 0$**.**
- **In the note's illustration:**

$$D(P \parallel P) = 0, \qquad D(P \parallel A) = 0.25, \qquad D(P \parallel B) = 1.25$$

- **Higher relative entropy means the estimated distribution deviates more from the actual distribution.**

Example 1.7

Consider a scenario where the true distribution is

$$P_{\textbf{dog}} = 1, \qquad P_{\textbf{cat}} = 0$$

For model A**,**

$$P'_{\textbf{dog}} = 0.1, \qquad P'_{\textbf{cat}} = 0.9$$

For model B**,**

$$P'_{\textbf{dog}} = 0.9, \qquad P'_{\textbf{cat}} = 0.1$$

Find the relative entropy using KL divergence.

For model A**:**

$$D(P \parallel P'_A) = P_{\textbf{dog}} \log_2 \frac{P_{\textbf{dog}}}{P'_{\textbf{dog}}} + P_{\textbf{cat}} \log_2 \frac{P_{\textbf{cat}}}{P'_{\textbf{cat}}}$$

$$= 1 \log_2 \frac{1}{0.1} + 0 \cdot \log_2 \frac{0}{0.9} = \log_2 10 = 3.32 \ \textbf{unit}$$

For model B**:**

$$D(P \parallel P'_B) = P_{\textbf{dog}} \log_2 \frac{P_{\textbf{dog}}}{P'_{\textbf{dog}}} + P_{\textbf{cat}} \log_2 \frac{P_{\textbf{cat}}}{P'_{\textbf{cat}}}$$

$$= 1 \log_2 \frac{1}{0.9} + 0 \cdot \log_2 \frac{0}{0.1} = 0.15 \ \textbf{unit}$$

So model B **is better because it has lower KL divergence.**

1.8.1 Mutual Information as Relative Entropy

For two dependent random variables (X, Y), mutual information can be written as the relative entropy between the joint distribution $P(x, y)$ and the product of the marginals $P(x)P(y)$:

Mutual Information from Relative Entropy

$$I(X;Y) = \sum_{x,y} P(x,y) \log_2 \frac{P(x,y)}{P(x)P(y)} = D\big(P(x,y) \,\|\, P(x)P(y)\big)$$

$$\begin{aligned}
I(X;Y) &= \sum_{x,y} P(x,y) \log_2 \frac{P(x)P(y \mid x)}{P(x)P(y)} \\
&= \sum_{x,y} P(x,y) \log_2 \frac{P(y \mid x)}{P(y)} \\
&= \sum_{x,y} P(x,y) \log_2 P(y \mid x) - \sum_{x,y} P(x,y) \log_2 P(y) \\
&= -H(Y \mid X) + H(Y)
\end{aligned}$$

So,

$$I(X;Y) = H(Y) - H(Y \mid X)$$

Also,

$$H(X,Y) = H(X) + H(Y \mid X)$$

Therefore,

$$H(Y \mid X) = H(X,Y) - H(X)$$

Substituting this into the previous expression,

$$I(X;Y) = H(Y) - H(X,Y) + H(X)$$

Thus the mutual information formula becomes

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

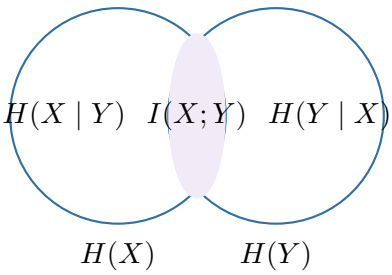

Figure 1.2: Mutual information as the overlap between $H(X)$ and $H(Y)$